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Adjacent-pair estimates for the Han–Xiong 1/2-conjecture

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Abstract
Han and Xiong defined Gaussian binomial coefficients with positive rational upper index and conjectured that the integer-exponent part is coefficientwise largest at the parameter $1/2$. Ono reduced the conjecture to the primitive parameters $1/(2m)$. We prove the primitive comparison for every $m \geq 2$ and $k \leq 6m + 2$, together with the additional case $(m, k) = (2, 15)$. The proof compares each primitive summand with two adjacent summands in the $1/2$-trace. All three local estimates are proved by finite product decompositions. As a noncomputational consequence, the conjecture holds for every positive rational parameter when $k \leq 15$. Moreover, the remaining primitive range for fixed k is reduced to $2 \leq m \leq \lfloor (k-3)/6 \rfloor$, with $(m, k) = (2, 15)$ removed.

Keywords Gaussian binomial coefficient · Integer trace · Coefficientwise positivity · Q-series

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1 Introduction

For $n \geq 0$, let

$$(z; q)_n := \prod_{j=0}^{n-1} (1 - zq^j), \qquad (z; q)_0 := 1.$$

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For a positive rational number r and an integer $k \geq 0$, Han and Xiong [2] defined

$$G_{r,k}(q) := \begin{bmatrix} r+k \\ k \end{bmatrix}_q := \frac{(q^{r+1};q)_k}{(q;q)_k}. \tag{1}$$

If $r = a/b$ is in lowest terms, then $G_{r,k}(q)$ is a formal power series in $q^{1/b}$. Its integer trace is obtained by keeping only the terms with integral exponents. We write this trace as

$$\mathcal{T}_{r,k}(q) := \mathrm{Tr}_{\mathbb{Z}}\big(G_{r,k}(q)\big).$$

For $F, G \in \mathbb{Q}[[q]]$, write

$$F(q) \succeq_q G(q) \iff [q^n]\big(F(q) - G(q)\big) \geq 0 \quad (n \geq 0).$$

Han and Xiong proposed the following conjecture.

Conjecture 1.1 *For every positive rational number r and every integer $k \geq 1$,*

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{r,k}(q).$$

Han and Xiong proved the conjecture for all positive rational r when $k \leq 3$, and for all k when r is a positive integer or a half-integer. They also verified the cases $r = 1/3$ and $r = 1/4$ for $k \leq 150$ by exact computation [2].

Ono [3] proved a support-dominance theorem and reduced Conjecture 1.1 to the primitive comparisons

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{1/(2m),k}(q) \qquad (m \geq 1,\ k \geq 1). \tag{2}$$

For fixed k, only $2 \leq m \leq \lfloor k/2 \rfloor$ can be nontrivial. Ono also verified every positive rational parameter for $k \leq 200$ by exact computation. Our result is complementary to this finite computation: it gives a uniform analytic region in (m, k) and consequently proves infinitely many cases with $k > 200$.

Our main result in this work is the following.

Theorem 1.2 *Let $m \geq 2$ and $k \geq 1$. Then*

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{1/(2m),k}(q)$$

whenever

$$k \leq 6m + 2.$$

The same comparison holds for $(m, k) = (2, 15)$.

The result has two immediate consequences.

Corollary 1.3 *Let $r = a/b$ be a positive rational number in lowest terms, and put*

$$L := \mathrm{lcm}(b, 2).$$

If $L = 2$, then

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{r,k}(q)$$

for every $k \geq 1$. If $L \geq 4$, then the same comparison holds for

$$k \leq 3L + 2.$$

It also holds when $L = 4$ and $k = 15$.

Corollary 1.4 *Conjecture 1.1 holds for every positive rational number r and every $1 \leq k \leq 15$.*

The proof of Theorem 1.2 uses the exact trace formula. For each summand indexed by ℓ in $\mathcal{T}_{1/(2m),k}$, we use the two summands indexed by

$$j = m\ell - 1 \qquad \text{and} \qquad j = m\ell$$

in $\mathcal{T}_{1/2,k}$. These pairs are disjoint because $m \geq 2$. All three local estimates are proved by finite product decompositions.

2 Preliminaries

2.1 The integer trace

Let $b \geq 1$, and let

$$F(q) = \sum_{n \geq 0} c_n q^{n/b} \in \mathbb{Q}[[q^{1/b}]].$$

Define

$$\mathrm{Tr}_{\mathbb{Z}}(F) := \sum_{j \geq 0} c_{bj} q^j. \tag{3}$$

Thus $\mathrm{Tr}_{\mathbb{Z}}(F)$ is the subseries of F supported on integral powers of q.

The finite q-binomial theorem [1, Chapter 3] is

$$(z;q)_k = \sum_{s=0}^{k} (-1)^s q^{\binom{s}{2}} \begin{bmatrix} k \\ s \end{bmatrix}_q z^s. \tag{4}$$

Substituting $z = q^{r+1}$ gives

$$(q^{r+1};q)_k = \sum_{s=0}^{k} (-1)^s q^{\binom{s+1}{2}+rs} \begin{bmatrix} k \\ s \end{bmatrix}_q. \tag{5}$$

The following exact trace formula follows from Han and Xiong's expansion [2] and is recorded in the form below by Ono [3].

Proposition 2.1 *Let $r = a/b$ be a positive rational number in lowest terms. Then*

$$\mathcal{T}_{a/b,k}(q) = \sum_{\substack{0\le s\le k\\ b\mid s}} (-1)^s q^{\binom{s+1}{2}+as/b} \frac{1}{(q;q)_s (q;q)_{k-s}} \tag{6}$$

$$= \sum_{j=0}^{\lfloor k/b\rfloor} (-1)^{bj} q^{\binom{bj+1}{2}+aj} \frac{1}{(q;q)_{bj}(q;q)_{k-bj}}. \tag{7}$$

Proof The exponent in the s-th term of (5) is integral if and only if as/b is integral. Since $\gcd(a,b) = 1$, this is equivalent to $b \mid s$. The factor $(q;q)_k^{-1}$ contains only integral powers of q, so the integer trace may be taken in the numerator. Since

$$\frac{1}{(q;q)_k}\begin{bmatrix} k \\ s \end{bmatrix}_q = \frac{1}{(q;q)_s (q;q)_{k-s}},$$

we obtain (6). Setting $s = bj$ gives (7). □

For $0 \le s \le k$, put

$$\Phi_{k,s}(q) := \frac{1}{(q;q)_s (q;q)_{k-s}}. \tag{8}$$

Each $\Phi_{k,s}(q)$ belongs to $\mathbb{Z}_{\ge 0}[[q]]$.

For the primitive parameter $1/(2\,m)$, Proposition 2.1 becomes

$$\mathcal{T}_{1/(2m),k}(q) = \sum_{\ell=0}^{\lfloor k/(2m)\rfloor} q^{2m^2\ell^2+(m+1)\ell} \Phi_{k,2m\ell}(q). \tag{9}$$

For $m = 1$, this is

$$\mathcal{T}_{1/2,k}(q) = \sum_{j=0}^{\lfloor k/2\rfloor} q^{2j(j+1)} \Phi_{k,2j}(q). \tag{10}$$

2.2 Ono's reduction

For a positive rational number r, let $\mathrm{den}(r)$ denote its denominator in lowest terms. Ono defines the even core

$$r^\flat := \frac{1}{\mathrm{lcm}(\mathrm{den}(r), 2)}. \tag{11}$$

His support-dominance theorem implies

$$\mathcal{T}_{r^\flat,k}(q) \succeq_q \mathcal{T}_{r,k}(q) \qquad (k \ge 1). \tag{12}$$

Consequently, Conjecture 1.1 is equivalent to the primitive family (2). For fixed k, only

$$2 \le m \le \left\lfloor \frac{k}{2} \right\rfloor \tag{13}$$

can be nontrivial [3].

3 The adjacent-pair estimate

For $a, d \geq 1$, write

$$[a]_{q^d} := \frac{1-q^{da}}{1-q^d}.$$

The next lemma compares one primitive summand in (9) with two adjacent summands in (10).

Lemma 3.1 *Let $m \geq 2$, $\ell \geq 1$, and put $M = m\ell$. If*

$$2M \leq k \leq (6m+4)\ell - 3, \tag{14}$$

then

$$q^{2M(M-1)}\Phi_{k,2M-2}(q) + q^{2M(M+1)}\Phi_{k,2M}(q) \\ \succeq_q q^{2m^2\ell^2+(m+1)\ell}\Phi_{k,2M}(q). \tag{15}$$

Proof *Factorization.* Set

$$g := (m-1)\ell, \qquad D := (3m+1)\ell, \qquad n := k - 2M.$$

Let

$$E := 2m^2\ell^2 + (m+1)\ell.$$

Then

$$E - 2M(M-1) = D, \qquad 2M(M+1) - E = g.$$

Hence the difference between the two sides of (15) is

$$q^{2M(M-1)}\left(\Phi_{k,2M-2}(q) - q^D(1-q^g)\Phi_{k,2M}(q)\right). \tag{16}$$

Put

$$K(q) := \frac{1}{(q;q)_{2M}(q;q)_{n+2}}.$$

Then the expression in parentheses in (16) is $K(q)P(q)$, where

$$P(q) := (1-q^{2M-1})(1-q^{2M}) \\ - q^D(1-q^g)(1-q^{n+1})(1-q^{n+2}). \tag{17}$$

Define

$$Q(q) := \frac{P(q)}{(1-q)^2(1-q^2)}. \tag{18}$$

Since $2M \geq 4$ and $n+2 \geq 2$, direct cancellation gives

$$K(q)(1-q)^2(1-q^2) = \frac{1}{(1-q^2)\prod_{d=3}^{2M}(1-q^d)\prod_{d=3}^{n+2}(1-q^d)} \in \mathbb{Z}_{\geq 0}[[q]]. \tag{19}$$

An empty product in (19) is understood to be 1. It is therefore enough to prove $Q(q) \in \mathbb{Z}_{\geq 0}[[q]]$.

Since

$$D = 3M + \ell, \qquad g = M - \ell,$$

we may write

$$Q(q) = \frac{(1-q^{2M-1})(1-q^{2M})}{(1-q)^2(1-q^2)} - q^{3M+\ell}C_n(q), \tag{20}$$

where

$$C_n(q) := \frac{(1-q^{M-\ell})(1-q^{n+1})(1-q^{n+2})}{(1-q)^2(1-q^2)}.$$

The series $C_n(q)$ is coefficientwise increasing with n. Indeed, if $n = 2r$, then

$$C_n(q) = [M-\ell]_q[2r+1]_q[r+1]_{q^2}, \qquad C_{n+1}(q) = [M-\ell]_q[2r+3]_q[r+1]_{q^2},$$

and if $n = 2r+1$, then

$$C_n(q) = [M-\ell]_q[2r+3]_q[r+1]_{q^2}, \qquad C_{n+1}(q) = [M-\ell]_q[2r+3]_q[r+2]_{q^2}.$$

Thus $C_{n+1}(q) \succeq_q C_n(q)$ in both cases.

The upper bound in (14) gives

$$n = k - 2M \leq 4M + 4\ell - 3.$$

Put

$$n_0 := 4M + 4\ell - 3.$$

Since $C_n(q) \preceq_q C_{n_0}(q)$ and $C_n(q)$ occurs in (20) with a negative sign, it follows that

$$Q(q) \succeq_q \widetilde{Q}(q),$$

where

$$\widetilde{Q}(q) := \frac{1}{(1-q)^2(1-q^2)}\big\{(1-q^{2M})(1-q^{2M-1}) - q^{3M+\ell}(1-q^{M-\ell})(1-q^{4M+4\ell-2}) \times (1-q^{4M+4\ell-1})\big\}. \tag{21}$$

It remains to prove that $\widetilde{Q}(q) \in \mathbb{Z}_{\geq 0}[[q]]$.

A direct expansion gives

$$\begin{aligned}
&(1-q)^2(1-q^2)\widetilde{Q}(q)\\
&= (1-q^{2M})(1-q^{2M-1})(1-q^{3M-\ell})\\
&\quad + q^{3M-\ell}(1-q^{2\ell})(1-q^{2M-1})(1-q^{2M})\\
&\quad + q^{4M}(1-q^{M+\ell-1})(1-q^{M+\ell})(1-q^{2M+2\ell-1})\\
&\quad + q^{7M+\ell-1}(1-q^{2\ell-1})(1-q^{2\ell})\\
&\quad + q^{7M+5\ell-1}(1-q^{M-\ell})(1-q^{4M+4\ell-2}).
\end{aligned} \tag{22}$$

After division by $(1-q)^2(1-q^2)$, every term on the right side has nonnegative coefficients. For the first term this follows from

$$\frac{(1-q^{2M})(1-q^{2M-1})(1-q^{3M-\ell})}{(1-q)^2(1-q^2)} = [M]_{q^2}[2M-1]_q[3M-\ell]_q.$$

For the second term, use the even factor $1-q^{2\ell}$. In the third term, one of the consecutive integers $M+\ell-1$ and $M+\ell$ is even. The fourth and fifth terms become, respectively,

$$q^{7M+\ell-1}\frac{[2\ell-1]_q[\ell]_{q^2}}{1-q}$$

and

$$q^{7M+5\ell-1}\frac{[M-\ell]_q[2M+2\ell-1]_{q^2}}{1-q}.$$

All factors in these expressions have nonnegative coefficients. Hence $\widetilde{Q}(q) \in \mathbb{Z}_{\geq 0}[[q]]$, and therefore $Q(q) \in \mathbb{Z}_{\geq 0}[[q]]$. Equations (16) and (19) now give (15). □

4 Endpoint estimates

The upper bound in Lemma 3.1 becomes $k \leq 6m+1$ when $\ell = 1$. The next lemma treats $k = 6m+2$.

Lemma 4.1 *For every $m \geq 2$,*

$$\begin{aligned}
&q^{2m(m-1)}\Phi_{6m+2,2m-2}(q) + q^{2m(m+1)}\Phi_{6m+2,2m}(q)\\
&\qquad \succeq_q q^{2m^2+m+1}\Phi_{6m+2,2m}(q).
\end{aligned} \tag{23}$$

Proof Let $\Delta_m(q)$ denote the difference between the two sides of (23). Using the common denominator

$$(q;q)_{2m}(q;q)_{4m+4},$$

we obtain

$$\Delta_m(q) = \frac{q^{2m^2-2m}}{(q;q)_{2m}(q;q)_{4m+4}} A_m(q), \tag{24}$$

where

$$\begin{aligned} A_m(q) = {} & (1-q^{2m-1})(1-q^{2m}) \\ & - q^{3m+1}(1-q^{m-1})(1-q^{4m+3})(1-q^{4m+4}). \end{aligned}$$

A direct expansion gives the identity

$$\begin{aligned} A_m(q) = {} & (1-q^{2m})(1-q^{2m-1})(1-q^{3m+1}) \\ & + q^{4m}(1-q^m)(1-q^{m+1})(1-q^{2m+1}) \\ & + q^{7m}(1-q)(1-q^2)(1-q^3) \\ & + q^{8m+2}(1-q)(1-q^2) \\ & + q^{7m+6}(1-q^{m-1})(1-q^{4m+2}). \end{aligned} \tag{25}$$

Each product on the right side of (25) cancels factors from the denominator in (24). For example,

$$\frac{(1-q^{2m})(1-q^{2m-1})(1-q^{3m+1})}{(q;q)_{2m}(q;q)_{4m+4}} \in \mathbb{Z}_{\geq 0}[[q]].$$

The same statement holds for the other four terms. After cancellation, each quotient is a product of reciprocal factors $(1-q^d)^{-1}$. All powers of q in front of these quotients are nonnegative. Hence every term obtained from (25) belongs to $\mathbb{Z}_{\geq 0}[[q]]$, and therefore $\Delta_m(q) \in \mathbb{Z}_{\geq 0}[[q]]$. □

The next lemma treats the additional case $(m, k) = (2, 15)$.

Lemma 4.2 *One has*

$$q^4\Phi_{15,2}(q) + q^{12}\Phi_{15,4}(q) \succeq_q q^{11}\Phi_{15,4}(q). \tag{26}$$

Proof Let $\Delta_*(q)$ denote the difference between the two sides of (26). With the common denominator $(q;q)_4(q;q)_{13}$,

$$\begin{aligned} \Delta_*(q) = \frac{1}{(q;q)_4(q;q)_{13}} \big\{ & q^4(q;q)_4 \\ & + q^5(1-q^2)(1-q^3)(1-q^4) \\ & + q^6(1-q^3)(1-q^4)(1-q^5) \\ & + q^{12}(1-q^2)(1-q^3)(1-q^4) \\ & + q^{16}(1-q)(1-q^3)(1-q^4) \\ & + q^{24}(1-q)(1-q^{12}) \big\}. \end{aligned} \tag{27}$$

To verify (27), multiply the original difference by $(q;q)_4(q;q)_{13}$. Its numerator is

$$q^4(1-q^3)(1-q^4) - q^{11}(1-q)(1-q^{12})(1-q^{13}),$$

and expansion gives the numerator displayed in (27).

Each product in braces cancels factors from $(q;q)_4(q;q)_{13}$. After cancellation, each quotient is a product of reciprocal factors $(1-q^d)^{-1}$, and therefore has nonnegative coefficients. It follows that $\Delta_*(q) \in \mathbb{Z}_{\geq 0}[[q]]$, which proves (26). □

5 Proof of the main theorem

Proof of Theorem 1.2 Write

$$L_j(q) := q^{2j(j+1)}\Phi_{k,2j}(q) \qquad \left(0 \leq j \leq \left\lfloor \frac{k}{2} \right\rfloor\right),$$

and

$$R_\ell(q) := q^{2m^2\ell^2+(m+1)\ell}\Phi_{k,2m\ell}(q) \qquad \left(0 \leq \ell \leq \left\lfloor \frac{k}{2m} \right\rfloor\right).$$

Then

$$\mathcal{T}_{1/2,k}(q) = \sum_j L_j(q), \qquad \mathcal{T}_{1/(2m),k}(q) = \sum_\ell R_\ell(q).$$

The terms with index zero agree:

$$L_0(q) = R_0(q) = \Phi_{k,0}(q).$$

For each $\ell \geq 1$ occurring in the second sum, we claim

$$L_{m\ell-1}(q) + L_{m\ell}(q) \succeq_q R_\ell(q). \tag{28}$$

The indices are valid because $2m\ell \leq k$.

Suppose first that $k \leq 6m+1$. Then

$$2m\ell \leq k \leq 6m+1 \leq (6m+4)\ell - 3,$$

so Lemma 3.1 proves (28) for every $\ell \geq 1$.

Now suppose $k = 6m+2$. Lemma 4.1 proves (28) for $\ell = 1$. If $\ell \geq 2$, then

$$(6m+4)\ell - 3 \geq 12m + 5 \geq 6m + 2 = k,$$

so Lemma 3.1 proves (28).

Finally, suppose $(m,k) = (2,15)$. Lemma 4.2 proves (28) for $\ell = 1$. For $\ell \geq 2$ with $4\ell \leq 15$,

$$(6m+4)\ell - 3 = 16\ell - 3 \geq 29 > 15,$$

so Lemma 3.1 proves the remaining local inequalities.

The pairs

$$P_\ell := \{m\ell - 1, m\ell\}$$

are pairwise disjoint. Indeed,

$$\min P_{\ell+1} - \max P_\ell = m - 1 \geq 1.$$

Let

$$J := \bigcup_{1 \leq \ell \leq \lfloor k/(2m) \rfloor} P_\ell.$$

Adding the local inequalities gives

$$\sum_{j \in J} L_j(q) \succeq_q \sum_{\ell=1}^{\lfloor k/(2m) \rfloor} R_\ell(q).$$

Every unassigned $L_j(q)$ has nonnegative coefficients. Therefore

$$\begin{aligned} \mathcal{T}_{1/2,k}(q) &= L_0(q) + \sum_{j \in J} L_j(q) + \sum_{\substack{1 \leq j \leq \lfloor k/2 \rfloor \\ j \notin J}} L_j(q) \\ &\succeq_q R_0(q) + \sum_{\ell=1}^{\lfloor k/(2m) \rfloor} R_\ell(q) \\ &= \mathcal{T}_{1/(2m),k}(q). \end{aligned}$$

This proves the theorem. □

6 Consequences

Proof of Corollary 1.3 Let $L = \operatorname{lcm}(b, 2)$. By (11),

$$r^\flat = \frac{1}{L}.$$

If $L = 2$, then $r^\flat = 1/2$, and (12) gives

$$\mathcal{T}_{1/2,k}(q) = \mathcal{T}_{r^\flat,k}(q) \succeq_q \mathcal{T}_{r,k}(q)$$

for every $k \geq 1$.

Suppose $L \geq 4$, and write $L = 2m$, so $m \geq 2$. If $k \leq 3L + 2$, then $k \leq 6m + 2$. Theorem 1.2 and (12) give

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{1/(2m),k}(q) = \mathcal{T}_{r^\flat,k}(q) \succeq_q \mathcal{T}_{r,k}(q).$$

If $L = 4$ and $k = 15$, then $m = 2$ and $r^\flat = 1/4$. The additional case of Theorem 1.2 gives

$$\mathcal{T}_{1/2,15}(q) \succeq_q \mathcal{T}_{1/4,15}(q) = \mathcal{T}_{r^\flat,15}(q).$$

Applying (12) completes the proof. □

Proof of Corollary 1.4 Let $L = \operatorname{lcm}(\operatorname{den}(r), 2)$. If $L = 2$, Corollary 1.3 applies for every k. If $L = 4$, it applies for $k \le 14$, and its additional clause gives $k = 15$. If $L \ge 6$, then $3L + 2 \ge 20$, so every $1 \le k \le 15$ satisfies the bound in Corollary 1.3. These cases exhaust all even values of L. □

Corollary 6.1 *For fixed k, a primitive comparison not covered by Theorem 1.2 can occur only when*

$$2 \le m \le \left\lfloor \frac{k-3}{6} \right\rfloor,$$

with $(m, k) = (2, 15)$ already covered.

Proof The inequality $k \le 6m + 2$ is equivalent to

$$m \ge \frac{k-2}{6}.$$

Thus a primitive parameter outside Theorem 1.2 must satisfy

$$6m + 2 < k,$$

or

$$m \le \frac{k-3}{6}.$$

Combining this with $m \ge 2$ gives the stated range. □

Remark 6.2 For $1 \le k \le 200$, the number of pairs in the range of Corollary 6.1 is

$$\sum_{k=1}^{200} \max\left(\left\lfloor \frac{k-3}{6} \right\rfloor - 1, 0\right) = 2{,}976.$$

Removing $(m, k) = (2, 15)$ leaves 2,975 pairs, compared with the 9,801 primitive pairs in Ono's original finite range.

7 The remaining problem

A complete proof of Conjecture 1.1 requires

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{1/(2m),k}(q)$$

for every $m \geq 2$ and $k \geq 1$. By Theorem 1.2, the remaining region is

$$k \geq 6m + 3,$$

except for the additional point $(m, k) = (2, 15)$.

The first unresolved primitive family is

$$\mathcal{T}_{1/2,k}(q) \succeq_q \mathcal{T}_{1/4,k}(q) \qquad (k \geq 1). \tag{29}$$

Han and Xiong define

$$H_k(q) := \sum_{j=1}^{\lfloor k/2 \rfloor} q^{2j(j+1)} \begin{bmatrix} k \\ 2j \end{bmatrix}_q - \sum_{\ell=1}^{\lfloor k/4 \rfloor} q^{\ell(8\ell+3)} \begin{bmatrix} k \\ 4\ell \end{bmatrix}_q.$$

Then

$$\mathcal{T}_{1/2,k}(q) - \mathcal{T}_{1/4,k}(q) = \frac{H_k(q)}{(q;q)_k}.$$

They conjecture $H_k(q) \succeq_q 0$ for $k \geq 19$ [2]. This is stronger than (29). Ono's cumulative-coefficient criterion [3] shows that it is enough to prove

$$\frac{H_k(q)}{1-q} \in \mathbb{Z}_{\geq 0}[[q]].$$

Lemma 3.1 uses two adjacent half-trace summands. For the first primitive summand $\ell = 1$, the present product decomposition reaches $k = 6m + 1$, and Lemma 4.1 treats $k = 6m + 2$. A proof beyond this range requires a different local comparison. One possibility is to compare a primitive summand with a longer block of half-trace summands.

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Declarations

Conflict of interest George E. Andrews is a member of the Editorial Board of *The Ramanujan Journal*. He will not participate in the peer-review process or in any editorial decision concerning this manuscript. Mohamed El Bachraoui has no relevant financial or non-financial interests to disclose.

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